

Assessment - Answer Key

Chapter 2: Diversity in the Living World

I. Choose the correct answer :

1. b) Lotus survives in water because its **waxy leaves float and do not get waterlogged**.
2. d) Correct answer is **all of these**, because camel, yak, and whale live in different habitats.
3. b) **Maize** is a monocot plant with one cotyledon.
4. b) A polluted pond with few fish shows **loss of biodiversity**.
5. c) Amphibians breathe through **lungs and moist skin**, which help them survive both on land and water.
6. d) Correct answer is **all of these**, as different animals use body parts for movement.

II. Match the following :

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|-----------------|------------------|
| 7. Monocot seed | – One cotyledon |
| 8. Dicot seed | – Two cotyledons |
| 9. Rhododendron | – Mountain plant |
| 10. Whale | – Aquatic mammal |
| 11. Frog | – Amphibian |

III. Answer the given questions as true or false:

12. False – Most fibrous-root plants have parallel venation, but exceptions exist.
13. False – Camels do not sweat; they conserve water.
14. False – Amphibians live on land and water.
15. True – Sacred groves protect forests and biodiversity.
16. True – Seeds help classify plants.

IV. Differentiate between the following:

17. Hot vs Cold Desert Camel

- Hot desert: One hump, survives with little water, short hair.
- Cold desert: Two humps, thick hair, adapted to cold.

18. Herbs, Shrubs, Trees

- Herbs: Small, soft stems, e.g., mint.
- Shrubs: Medium, woody stems, e.g., rose.
- Trees: Tall, woody trunk, e.g., neem.

V. Answer the following questions in brief:

19. Rhododendrons have small leaves to reduce water loss and protect from cold.
20. Biodiversity is not the same everywhere because climate and resources differ, like deserts have fewer species while rainforests have more.
21. Frogs have lungs and moist skin for breathing and webbed feet for swimming, making them fit for both water and land.
22. Sacred groves are patches of forests protected by people for cultural reasons, helping biodiversity survive.

VI. Answer the following questions in detail:

23. Classifying Plants in a School Garden

If I were a researcher classifying plants in a school garden, I would use features like height, venation, and roots. First, I would divide them into herbs, shrubs, and trees based on their stem and height. Then I would check their leaves: if the veins are parallel, the plant is a monocot, and if the veins are net-like, it is a dicot. Finally, I would examine their roots: fibrous roots belong to monocots, while taproots belong to dicots. This method would give a clear, systematic record of the plants and make the study easier.

24. Aquatic vs Terrestrial Habitats

Aquatic habitats are water-based, and plants and animals here adapt to survive in water. For example, lotus plants have waxy leaves that float, and fish have gills for breathing. Terrestrial habitats are land-based, where organisms adjust to hot, cold, or dry conditions. Cactus has spines instead of leaves, and camels have humps to store fat for survival in deserts. Both habitats present challenges, and organisms evolve different features to survive. This comparison shows the diversity of life forms in different environments.

25. Grouping and Biodiversity

Grouping plants and animals helps us understand biodiversity in a clear way. If we randomly study each organism, it becomes confusing, but grouping shows patterns and similarities. For example, grouping plants as monocots and dicots links seeds, roots, and leaves together. Grouping animals by movement, like flying, swimming, or crawling, shows how their bodies are adapted. By grouping, scientists can classify endangered species and plan conservation efforts. It also helps students and communities to appreciate the richness of biodiversity and the need to protect it.